

The Odu-256 Training Curriculum: A 256-State Product Taxonomy for Structured Language-Model Training and Evaluation

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Abstract

Curriculum learning [Bengio 2009] and structured evaluation taxonomies [Hendrycks 2021; Liang 2023; BIG-Bench 2023] are two of the more robust levers for improving the sample efficiency and the interpretability of language-model training. Curricula need an ordering; evaluation suites need strata. Both need a state space. The dominant designs use either task-level partitions (one cell per benchmark subject) or skill-level partitions (one cell per learned ability), with cell counts in the tens to low hundreds.

We propose **Odu-256**: a 256-cell training and evaluation taxonomy derived from the 256 Odu of the Ifá divination tradition (16 Meji $\times$ 16 Meji), reinterpreted as a *product* of two 16-state archetypal masks over the TERA register (Temporal, Emotional, Rational, Archetypal). Each of the 256 cells is identified with an ordered pair (M_upper, M_lower) of named 4-bit archetypes; the diagonal 16 cells correspond to single-archetype resonance states and the 240 off-diagonal cells correspond to ordered archetypal transitions.

We treat Odu-256 strictly as a **structured discrete state space**, not as a divinatory instrument. The contribution is mathematical and methodological: (i) the cell structure admits closed-form factorization into TERA components and into upper/lower marginals; (ii) the cells form a natural curriculum-ordering by archetypal complexity (popcount of the 8-bit code); (iii) the taxonomy yields a stratified evaluation harness with provable coverage guarantees on TERA-defined subspaces. We describe the data-bucketing protocol, the sampling and re-weighting procedures, the evaluation stratification, and integration with the Vortex-keyed router [companion preprint 01-vortex-keyed-moe-routing]. Empirical validation is in progress; results will be reported in v1.1.

Code: Catalog and runtime live in the AsAManThinks platform (`packages/aamt-foundations/oracle-256-car` and `packages/vortex-runtime/`) and the MaiaM Alchemist project (`packages/code-corpus/`).

1. Introduction

The two-decade arc from MNIST [LeCun 1998] to MMLU [Hendrycks 2021] to HELM [Liang 2023] to BIG-Bench [BIG-Bench 2023] tracks an increasing appreciation for **structured stratification** of training and evaluation data. The earliest benchmarks were monolithic; the current generation is hierarchical, multi-axis, and explicitly typed.

Curriculum learning [Bengio 2009; Soviany 2022] runs the same argument on the training side: presenting examples in an order informed by *difficulty* or *type* improves both convergence speed and generalization in many regimes. The choice of ordering — what counts as a stage, what counts as harder — is the substantive design decision. Existing curricula use proxy signals (loss, perplexity, data source), task-level partitions (BIG-Bench’s task hierarchy), or human-curated skill axes (HELM’s scenarios).

We propose a different design: an explicit, **product-structured** 256-cell taxonomy in which each cell is identified by an ordered pair of named 4-bit archetypal masks. The 4-bit base — the **Meji** — indexes the $2^4 = 16$ possible activation patterns of the four TERA dimensions [Whitehead 2024].

The product of two Mejis yields the 256-state **Odu** lattice, which we adopt as the curriculum and evaluation taxonomy.

The taxonomy is grounded in the 256 Odu of the Ifá tradition (Iyanifa Faye’s “Cosmic Web of Life” diagram). We make our use of this grounding explicit: we are not using Odu as divination, and we are not making claims about divinatory efficacy. We are using the *structure* — 16 single-archetype states tensored with themselves to yield 256 ordered pairs — as a pre-built, culturally-validated 256-element discrete state space with a coherent semantic labeling. The labeling is documentation; the architecture would function with purely numeric labels. Section 2.4 discusses the cultural-lineage attribution and the user-facing naming policy.

Our contributions:

1. The **Odu-256 product taxonomy** as a 256-cell training and evaluation state space, with the 16-Meji base $\times$ 16-Meji product factorization.
2. A **closed-form factorization** of Odu cells into TERA components (upper and lower) admitting marginalization to coarser strata ($16 \times 16 = 256$, $16 = 16$, $2^4 = 16$ dimension-wise).
3. A **data bucketing protocol** for assigning training/evaluation examples to Odu cells via a soft Meji distribution induced by the per-example TERA vector.
4. A **curriculum ordering** by archetypal complexity (8-bit popcount) with empirical justification for staged training.
5. An **evaluation stratification harness** providing coverage guarantees across TERA subspaces, archetype-marginals, and full Odu cells.
6. Integration with the **Vortex-keyed router** [companion preprint] when both router and curriculum address the same 16-Meji frame.

The remainder proceeds as follows. Section 2 covers related work on curriculum learning, taxonomies, and cultural-data attribution. Section 3 defines the Meji and Odu state spaces. Section 4 describes the data-bucketing protocol. Section 5 specifies the curriculum. Section 6 specifies the evaluation harness. Section 7 outlines the proposed empirical program. Section 8 enumerates limitations. Section 9 concludes.

2. Background and Related Work

2.1 Curriculum learning

Curriculum learning [Bengio 2009] frames training as a sequence of example distributions of increasing difficulty. Soviany et al. (2022) survey the field, distinguishing **data-level** curricula (which examples), **task-level** curricula (which tasks), and **model-level** curricula (which sub-modules are trainable). Self-paced learning [Kumar 2010] makes the difficulty signal model-derived rather than pre-specified. Recent applications to large language models include multi-stage pretraining [Du 2022; Gemma Team 2024] and instruction curriculum [Wei 2022].

The Odu-256 curriculum is a **structured data-level curriculum**: the ordering is defined by the popcount of the 8-bit Odu code, which is a proxy for archetypal complexity (how many TERA dimensions are active in upper-and-lower combined). The ordering is *deterministic* — unlike self-paced learning — and *content-grounded* — unlike random or source-based shuffles.

2.2 Evaluation taxonomies

MMLU [Hendrycks 2021] organizes 57 academic subjects into 4 super-categories. HELM [Liang 2023] proposes a multi-axis scenario matrix with explicit coverage targets. BIG-Bench [BIG-Bench

2023] provides 200+ tasks with metadata fields for skills, behaviors, and input types. Holistic frameworks such as BIG-Bench Hard [Suzgun 2023] re-stratify by difficulty post-hoc.

The Odu-256 taxonomy differs from these in three respects:

1. **Product structure.** The 256 cells are not flat labels; they are ordered pairs from a 16-element base. This admits marginalization to coarser strata and re-aggregation along TERA dimensions.
2. **Determinism.** Cell assignment is a closed-form function of the example’s TERA vector. There is no learned classifier in the loop, no human annotator subjectivity beyond the TERA-anchoring step.
3. **Coverage by construction.** Every cell exists *a priori*; the only empirical question is how the corpus is distributed across cells. Sparse cells are a coverage gap, not an absence of category.

2.3 Discrete archetypal taxonomies

The use of small, fixed sets of archetypal categories for cognitive or affective state is established in psychology (Plutchik’s 8-emotion wheel [Plutchik 1980]; Ekman’s basic emotions [Ekman 1992]) and in cross-cultural symbol systems (the I Ching’s 64 hexagrams [Leibniz 1703 binarization]; the 16-personality Myers-Briggs framework [Myers 1962]). The Ifá tradition’s 16 Meji and 256 Odu form one such system, with the distinguishing property that the product structure ($\text{Meji} \times \text{Meji} = \text{Odu}$) is intrinsic to the tradition’s own combinatorial logic rather than imposed post-hoc.

We adopt the Odu-256 structure for three engineering reasons: (i) 256 is a useful cardinality — large enough to support fine-grained stratification, small enough to admit per-cell evaluation budgets in practice; (ii) the product factorization yields closed-form marginal distributions for free; (iii) the 16-Meji base aligns directly with the 16-expert Vortex-keyed router, enabling joint curriculum/architecture experiments.

2.4 Cultural lineage and naming policy

The 256 Odu lattice originates in the Ifá divination tradition of the Yoruba people. Our use of it is structural, not divinatory; the mathematical content is the 4-bit $\times$ 4-bit product. We follow a strict naming policy to honor lineage without appropriating practice:

- **Internal identifiers** use numeric Meji ids (0..15) and Odu pair ids (“upper-lower”). All telemetry, logs, and runtime code use the numeric form. There is no operational dependence on the cultural name.
- **Ifá lineage names** (Eji-Ogbe, Oyeku Meji, ..., Ofun Meji) are retained in the JSON metadata for attribution. They are *never* surfaced in user-facing UI without practitioner review.
- **User-facing names** use an AAMT-native vocabulary (Apex, Hollow, Heart-Mind, Threshold, Bloom, Root, Compass, Echo, Forge, Seed, Blade, Tide, Stream, Lens, Weave, Now) chosen to be brand-consistent, English-rooted, and non-appropriative. The mapping table is stable across all surfaces.

The reference catalog (`packages/aamt-foundations/yare-vortex-mathematics.json`) carries the attribution explicitly: “Cosmological foundation: Odu Ifá tradition (Iyanifa Faye). User-facing names use the AAMT convention; Ifá lineage names retained in metadata for attribution.” A regeneration validator enforces that the lineage tokens do not leak into user-facing strings.

3. The Meji and Odu State Spaces

3.1 The Meji: 16 single-archetype masks

The base state space is the 16-element **Meji** lattice. Each Meji is a 4-bit code indexed by the four TERA dimensions:

$$M \in \{0, 1\}^4 = (T, E, R, A)$$

with bit assignments $T = \text{line 4}$ (most significant), $E = \text{line 3}$, $R = \text{line 2}$, $A = \text{line 1}$. Each bit indicates whether the corresponding TERA dimension is *active* (1, open) or *latent* (0, closed) in that archetype. There are $2^4 = 16$ such masks. The 16 are labeled (Table 1) with both their Ifá lineage names (for attribution) and their AAMT- native names (for user-facing surfaces). The bit assignment is an AAMT convention chosen so that the saturation pattern (1111) maps to the maximal-expansion archetype and the null pattern (0000) maps to the minimal-activation archetype, matching the polar archetypes of the cosmology.

Table 1: The 16 Meji. (Abbreviated; full table in Appendix A.)

id	binary	AAMT name	tera_active	polarity
0	1111	Apex	T, E, R, A	F-pure
1	0000	Hollow	(none)	M-pure
2	0110	Heart-Mind	E, R	balanced
3	1001	Threshold	T, A	balanced
4	1100	Bloom	T, E	F-leaning
5	0011	Root	R, A	M-leaning
6	1010	Compass	T, R	alternating
7	0101	Echo	E, A	alternating
8	1110	Forge	T, E, R	F-leaning
9	0001	Seed	A	M-leaning
10	1011	Blade	T, R, A	F-leaning
11	0100	Tide	E	M-leaning
12	1101	Stream	T, E, A	F-leaning
13	0010	Lens	R	M-leaning
14	0111	Weave	E, R, A	M-leaning
15	1000	Now	T	F-leaning

3.2 The Odu: 256 ordered Meji pairs

The full state space is the **Odu** lattice, the ordered tensor product $\text{Meji} \times \text{Meji}$:

$$\text{Odu} = M_{\text{upper}} \otimes M_{\text{lower}} \in \{0, 1\}^4 \times \{0, 1\}^4$$

with cardinality $16 \times 16 = 256$ and 8-bit binary encoding ($T_F, E_F, R_F, A_F, T_M, E_M, R_M, A_M$). The 256 cells decompose into:

- **16 diagonal cells** ($M_{\text{upper}} = M_{\text{lower}}$) — *resonance* states, in which the upper and lower masks agree. The diagonal is the locus of single-archetype pure-mode states.
- **240 off-diagonal cells** ($M_{\text{upper}} \neq M_{\text{lower}}$) — *transition* states, in which the upper mask differs from the lower mask. The ordered pair $(M_{\text{upper}}, M_{\text{lower}})$ is distinct from $(M_{\text{lower}}, M_{\text{upper}})$; direction matters.

3.3 Closed-form factorization

The product structure yields three natural marginalizations:

Upper marginal. Summing over lower Mejis gives the 16-cell distribution over upper Mejis only:

$$p_U(M_u) = \sum_{M_l} p(M_u, M_l)$$

Lower marginal. Symmetrically,

$$p_L(M_l) = \sum_{M_u} p(M_u, M_l)$$

TERA-dimension marginals. Summing over all Mejis with a given bit in a given position gives the 2-cell distribution per TERA dimension per side:

$$p_{T_F=1} = \sum_{M_u: T_F(M_u)=1} p_U(M_u)$$

and similarly for E, R, A in upper and lower. Eight such marginals (2 sides $\times$ 4 dimensions) provide a low-dimensional summary of corpus coverage.

3.4 Soft Meji distribution from a TERA vector

The bridge from a continuous-valued TERA vector $\tau = (T, E, R, A) \in [0,1]^4$ to a distribution over the 16 Mejis is the **soft Meji** distribution:

$$p(M_k | \tau) = \prod_{d \in \{T, E, R, A\}} \tau_d^{b_d(k)} \cdot (1 - \tau_d)^{1-b_d(k)}$$

where $b_d(k)$ is the d -th bit of k . This is the joint probability that each TERA dimension lies in the half-space specified by the k -th mask, under independence. The 16 weights sum to 1 by construction.

The induced **soft Odu distribution** is the outer product of two independent soft-Meji draws (one for upper, one for lower):

$$p((M_u, M_l) | \tau_F, \tau_M) = p(M_u | \tau_F) \cdot p(M_l | \tau_M)$$

where τ_F and τ_M are the TERA vectors of the upper (F-phase) and lower (M-phase) profiles. In the simplest setting $\tau_F = \tau_M = \tau$; in the breath-cycle setting they differ.

This is the same closed-form distribution used in soft-Vortex routing [companion preprint, §4.2]; it is reproduced here as the *bucketing function* for the curriculum rather than as a router. The reference implementation lives at `packages/vortex-runtime/src/index.ts:mejiDistribution`.

4. Data Bucketing Protocol

4.1 Per-example TERA labels

Each training or evaluation example is labeled with a TERA vector $\tau \in [0,1]^4$. Three sources are supported:

1. **Source-derived.** For corpora with explicit metadata (genre, modality, register), a deterministic rule maps metadata to TERA. For example, narrative fiction $\rightarrow$ A high, R low; technical documentation $\rightarrow$ R high, A low; live-time chat $\rightarrow$ T high.
2. **Model-derived.** A pre-trained TERA classifier (a four-output sigmoid head on a frozen base model, trained on a labeled subset) produces τ for each example. The labeled subset can be small ($\sim 10K$ examples per dimension); the rest is unsupervised.
3. **Heuristic.** For lightweight labeling, simple textual features (verb tense $\rightarrow$ T; affect-word density $\rightarrow$ E; logical-connective density $\rightarrow$ R; metaphor density $\rightarrow$ A) suffice for coarse bucketing.

The bucketing protocol is agnostic to label source provided $\tau \in [0,1]^4$ is available per example.

4.2 Cell assignment

Given an example with TERA label τ , the example is assigned to Odu cells using one of three modes:

- **Hard assignment.** The example is assigned to the single Odu cell (M_u, M_l) with the highest soft probability. This is the cell- per-example mode; total mass per cell = (# examples in that cell).
- **Soft assignment.** The example contributes weight $p((M_u, M_l) | \tau)$ to every cell. This is the fractional-mass mode; total mass per cell = sum of per-example weights.
- **Sampled assignment.** k draws are taken from the soft distribution and the example is assigned to each sampled cell with weight $1/k$. This is a stochastic approximation to soft assignment with bounded per-example storage.

We use soft assignment in the reference implementation. The fractional weights are stored sparsely; a typical example contributes non-zero mass to ~ 4 -8 cells out of 256.

4.3 Cell statistics

For each of the 256 cells we maintain a small set of statistics: example count (or soft-mass), token count, mean per-token loss during training, and the empirical distribution of source metadata (genre, register, source corpus) within the cell. These statistics are surfaced in a dashboard and used as the input to the curriculum ordering (Section 5) and the re-weighting policy (Section 4.4).

4.4 Re-weighting

Cells with low example count are upsampled during training to ensure coverage. The reference policy is **inverse-frequency** with a floor:

$$w(M_u, M_l) = \min(w_{\max}, \frac{\bar{n}}{\max(n(M_u, M_l), n_{\min})})$$

where $n(M_u, M_l)$ is the soft-mass of the cell, $\bar{n}$ is the mean over cells, $n_{\min}$ is a floor preventing infinite weight, and $w_{\max}$ is a cap preventing pathological upsampling. We use $n_{\min} = 100$

examples, $w_max = 10$.

Re-weighting is applied at the sampling stage (mini-batch construction draws examples with probability proportional to cell weight) rather than at the loss stage.

5. The Curriculum

5.1 Stage ordering

The Odu-256 curriculum stages examples by **archetypal complexity**, defined as the popcount of the 8-bit Odu code:

$$c(M_u, M_l) = \text{popcount}(M_u) + \text{popcount}(M_l) \in \{0, 1, \dots, 8\}$$

Cells are partitioned into 9 stages by complexity. Stage 0 contains the single cell (Hollow, Hollow) — both masks fully closed. Stage 8 contains the single cell (Apex, Apex) — both masks fully open. Stages in between contain the multinomially-distributed remainder.

The stage cardinalities follow the binomial coefficients:

stage	upper popcount $\times$ lower popcount cells	count
0	1×1	1
1	$(1 \times 4) + (4 \times 1)$	8
2	$(1 \times 6) + (4 \times 4) + (6 \times 1)$	28
3	$(1 \times 4) + (4 \times 6) + (6 \times 4) + (4 \times 1)$	56
4	$(1 \times 1) + (4 \times 4) + (6 \times 6) + (4 \times 4) + (1 \times 1)$	70
5	(symmetric to stage 3)	56
6	(symmetric to stage 2)	28
7	(symmetric to stage 1)	8
8	1×1	1

The total is $1 + 8 + 28 + 56 + 70 + 56 + 28 + 8 + 1 = 256$, as expected.

The proposed schedule starts with stages 0-2 (the low-activation cells), progresses through stages 3-5 (the median-activation cells), and ends with stages 6-8 (the high-activation cells). The intuition is that low-popcount cells correspond to single-dimension-dominant examples (Hollow $\leftrightarrow$ minimal-activation, Lens $\leftrightarrow$ rational-only, etc.) — which we hypothesize are easier to learn coherent representations for than mixed-dimension cells.

We do *not* claim this ordering is optimal. We claim it is defensible, deterministic, and reproducible, and we explicitly invite ablation against the natural alternative orderings (uniform; reverse; by empirical loss; by source-corpus difficulty).

5.2 Stage transitions

Following Bengio (2009) we use a **smooth** stage transition rather than a hard cut. At step t , the active cell distribution is:

$$\pi_t(M_u, M_l) \propto w(M_u, M_l) \cdot \mathbb{1}[c(M_u, M_l) \leq s(t)] + \alpha \cdot \mathbb{1}[c(M_u, M_l) > s(t)]$$

where $s(t)$ is the active stage at step t (advancing on a schedule) and α is a small mixing weight (we use $\alpha = 0.1$) that keeps non- active stages reachable for downstream calibration.

5.3 Within-stage sampling

Within each stage, examples are sampled by inverse-frequency weighting over the cells in that stage (Section 4.4). This decouples the curriculum’s *what-stage-now* signal from the corpus’s *what-cell-is-rare* signal: rare cells within the active stage are upsampled regardless of the overall corpus distribution.

5.4 Convergence and termination

The curriculum terminates when all 9 stages have been visited and the per-cell loss converges within a target tolerance. The termination criterion is per-cell rather than global: a stage is “done” when the loss in every cell within it has plateaued. This prevents premature exit from a stage in which a rare cell has not yet been adequately trained.

6. Evaluation Stratification

6.1 Per-cell evaluation

The 256 Odu cells serve as the evaluation strata. Every benchmark example is bucketed (Section 4.2) and per-cell metrics are reported alongside the global metric. This provides immediate visibility into coverage gaps and uneven capability.

For benchmarks small enough that 256-way stratification is sparse, we report at coarser granularity: by stage (9 strata), by upper- marginal (16 strata), by lower-marginal (16 strata), or by TERA dimension (8 strata: T-active, T-latent, E-active, E-latent, ...).

6.2 Coverage guarantees

A benchmark is said to have **complete Odu coverage** if every cell has at least k examples, for a chosen minimum k . For $k = 16$, complete coverage requires $256 \times 16 = 4096$ examples; for $k = 64$, 16,384 examples. The Odu-256 coverage matrix is reported as a 16×16 heatmap alongside the headline metric, providing a single-glance audit of stratification.

Complete coverage at $k = 1$ is achievable by construction for any benchmark of size ≥ 256 via stratified re-sampling. Whether the post-sampled distribution is representative of the underlying phenomenon being measured is, of course, a substantive question; the coverage guarantee is structural, not semantic.

6.3 Comparison to existing stratifications

MMLU’s 4 super-categories and 57 subjects provide a coarse 57-cell taxonomy with strong subject coherence but weak product structure. HELM’s scenario matrix offers richer axes but cell counts vary widely. BIG-Bench’s task metadata supports many slicings but has no canonical 256-cell view. The Odu-256 stratification differs in providing a *fixed* 256-cell view that is closed-form factorizable. This trades semantic richness per cell (an Odu cell is “Apex $\times$ Heart-Mind transition,” which is abstract) for compositional structure (every cell is identified by a typed product of TERA masks). The two are complementary; the Odu stratification is intended as an *additional* axis, not a replacement.

6.4 Routing-aware evaluation

For models trained with the Vortex-keyed router [companion preprint], the Odu cell of an example can be compared to the cell visited by the router during inference. Discrepancy between the corpus-assigned cell and the routing-assigned cell is a routing-error signal that is not available with any other stratification scheme. This is the joint curriculum/architecture experiment we treat as the primary empirical target.

7. Proposed Empirical Program

7.1 Base setup

- **Base model:** Gemma-2-2B [Gemma Team 2024].
- **Pretraining corpus:** a held-out slice of C4 [Raffel 2020] with per-example TERA labels from a frozen TERA classifier.
- **Curriculum schedule:** linear advancement through the 9 stages over the first 80% of training, followed by uniform-cell sampling for the final 20%.

7.2 Comparisons

- **Random shuffle baseline.** Standard uniform-random batching.
- **Reverse curriculum.** Stages $8 \rightarrow 0$; tests whether the proposed ordering matters or only the *staging* matters.
- **Source-difficulty curriculum.** Stages defined by source-corpus perplexity from the base model; tests whether Odu-derived ordering outperforms a loss-derived one.
- **Odu curriculum** (this work).

7.3 Tasks

- **Pretraining perplexity** on a held-out C4 slice.
- **MMLU** [Hendrycks 2021], reported overall and per-Odu-cell.
- **HellaSwag** [Zellers 2019], same.
- **TruthfulQA** [Lin 2022], same.
- **BIG-Bench Lite** [BIG-Bench 2023], reported per-stage.

7.4 Curriculum-specific evaluations

- **Per-cell convergence trace.** Does the loss in each cell plateau in stage order, or does it converge globally with no stage signature?
- **Cell coverage.** Do all 256 cells receive non-trivial training mass, or does the corpus concentrate?
- **Curriculum transfer.** Does pretraining with Odu curriculum improve downstream stratified evaluation more than uniform pretraining?

7.5 Joint curriculum/router ablation

The most distinctive experiment: train a Vortex-keyed router model [companion preprint] with the Odu curriculum, where the router’s 16 experts correspond to the 16 Meji. Measure:

- **Router-corpus alignment.** Does the router learn to route examples to the Meji their TERA labels were sampled from? Match rate vs. random expectation (1/16) is the headline.

- **Stage-conditioned routing entropy.** Does routing entropy drop as the curriculum advances (the model has more confident archetype assignments) or stay constant?

8. Limitations

No headline numbers in v1.0. The empirical case is preliminary. We are running the proposed evaluations and will report in v1.1.

TERA labels are a prerequisite. The bucketing protocol requires per-example TERA vectors. For small labeled subsets a frozen TERA classifier suffices; for very large corpora the classifier inference cost is non-trivial. Heuristic TERA labels are cheaper but coarser.

256 cells is small at large corpus scale. A 1B-token corpus distributed across 256 cells averages 4M tokens per cell, which is sufficient. A 100B-token corpus averages 400M per cell, which is arguably overkill. Whether the stratification continues to add signal at frontier scale is an open empirical question; finer stratifications (1024 cells via 5-bit Mejis, 4096 cells via 6-bit Mejis) are a natural extension but lose the cultural-lineage grounding.

Stage ordering is hypothesized, not derived. The popcount-based ordering is defensible but not optimal. A self-paced variant that re-orders stages based on model-derived difficulty signals would be a natural follow-up.

Cultural grounding may invite mis-reading. Despite the explicit naming-policy machinery (Section 2.4), readers may infer that the Odu structure is being used divinatorily. It is not. We have made this explicit in the paper, in the catalog metadata, and in the runtime code (which uses numeric ids exclusively at the operational layer).

Independence assumption in the soft distribution. The soft Meji formula (Section 3.4) assumes TERA dimensions are independent. They are typically *correlated* in natural data (a high-T example tends to also be high-E in many corpora). The soft distribution underestimates joint probabilities; corrections via a learned 4-bit joint factor are straightforward but add complexity.

Cell labels are documentation, not architecture. A skeptical reader may note that the named cells (Apex, Hollow, ..., or Eji-Ogbe, Oyeku Meji, ...) do not by themselves produce curriculum benefit. We agree: the value is the 256-cell *structure* and the closed-form *factorization*. The names are a convenience for the human auditor inspecting cell statistics; the structure is what does the work.

9. Discussion and Future Work

9.1 Connection to product taxonomies

The Odu-256 lattice is one instance of a general design pattern: *build the evaluation taxonomy as a product of small, named bases*. The benefit is closed-form factorization and explicit marginal distributions. The cost is interpretive abstraction at the cell level.

We expect this pattern to generalize. A Plutchik $\times$ Plutchik product [Plutchik 1980] yields a 64-cell emotion-transition lattice. A MBTI $\times$ MBTI product yields a 256-cell personality-pair lattice. The Odu-256 instance is distinguished by the cultural validation of the specific 16-base and by its alignment with the TERA register and the Vortex-keyed router.

9.2 Connection to multi-axis evaluation

HELM [Liang 2023] argues for explicit multi-axis evaluation matrices. The Odu-256 stratification is one such matrix, with two named axes (upper Meji, lower Meji), each with four further sub-axes (the TERA bits). Per-cell, per-marginal, and per-TERA-dimension metrics are all available from the same bucketing pass.

9.3 Future work

- **Per-stage capacity scaling.** Allocate FLOPs per stage in proportion to stage cell count (stage 4 has 70 cells; stage 0 has 1). Test whether the natural multinomial cell count is the right FLOP allocation prior.
- **Cross-cultural validation.** Compare Odu-256 stratification to a parallel I-Ching-64 $\times$ I-Ching-64 = 4096 lattice and to a smaller Plutchik $\times$ Plutchik = 64 lattice on the same corpus.
- **Adaptive cell pruning.** Identify cells with persistently low empirical mass and either upsample aggressively or merge them with neighbors in Hamming distance.
- **Per-cell evaluation budgets.** Allow per-cell evaluation example counts to grow under coverage-guarantee constraints ($k \geq 16$ in every cell).

10. Conclusion

We presented the Odu-256 training curriculum: a 256-cell product taxonomy over the 16-Meji TERA-mask base, used as both a curriculum ordering and an evaluation stratification. The cells are defined by closed-form factorization of a continuous-valued TERA label, the ordering is by archetypal complexity (8-bit popcount), and the stratification provides coverage guarantees across TERA subspaces, archetype-marginals, and the full 256-cell lattice.

The contribution is methodological: a pre-built, structured 256- element state space with cultural-lineage attribution and explicit factorization properties, usable as a drop-in stratification axis alongside MMLU, HELM, and BIG-Bench. Joint experiments with the Vortex-keyed router are the primary empirical target; results will appear in v1.1.

References

- Bengio, Y., Louradour, J., Collobert, R., Weston, J. (2009). *Curriculum Learning*. ICML.
- BIG-Bench Authors (2023). *Beyond the Imitation Game: Quantifying and Extrapolating the Capabilities of Language Models*. TMLR.
- Du, N. et al. (2022). *GLaM: Efficient Scaling of Language Models with Mixture-of-Experts*. ICML.
- Ekman, P. (1992). *An argument for basic emotions*. Cognition and Emotion.
- Gemma Team (2024). *Gemma 2: Improving Open Language Models at a Practical Size*. arXiv:2408.00118.
- Hendrycks, D., Burns, C., Basart, S., Zou, A., Mazeika, M., Song, D., Steinhardt, J. (2021). *Measuring Massive Multitask Language Understanding*. ICLR.
- Kumar, M. P., Packer, B., Koller, D. (2010). *Self-Paced Learning for Latent Variable Models*. NeurIPS.
- LeCun, Y., Bottou, L., Bengio, Y., Haffner, P. (1998). *Gradient- Based Learning Applied to Document Recognition*. Proc. IEEE.

- Leibniz, G. W. (1703). *Explication de l’arithmétique binaire*. Mémoires de l’Académie Royale des Sciences.
- Liang, P. et al. (2023). *Holistic Evaluation of Language Models (HELM)*. TMLR.
- Lin, S., Hilton, J., Evans, O. (2022). *TruthfulQA: Measuring How Models Mimic Human Falsehoods*. ACL.
- Myers, I. B. (1962). *The Myers-Briggs Type Indicator: Manual*. Educational Testing Service.
- Plutchik, R. (1980). *A general psychoevolutionary theory of emotion*. In *Emotion: Theory, Research, and Experience*, Vol. 1. Academic Press.
- Raffel, C. et al. (2020). *Exploring the Limits of Transfer Learning with a Unified Text-to-Text Transformer (C4)*. JMLR.
- Soviany, P., Ionescu, R. T., Rota, P., Sebe, N. (2022). *Curriculum Learning: A Survey*. IJCV.
- Suzgun, M. et al. (2023). *Challenging BIG-Bench Tasks and Whether Chain-of-Thought Can Solve Them (BIG-Bench Hard)*. ACL Findings.
- Wei, J. et al. (2022). *Finetuned Language Models Are Zero-Shot Learners*. ICLR.
- Whitehead, W. C. Jr. (2024). *AAMT Foundations: TERA, Vortex, and HeartScale*. AsAManThinks technical report.
- Whitehead, W. C. Jr. (2026). *Vortex-Keyed Mixture-of-Experts Routing: A Deterministic, Interpretable Gating Primitive*. AsAManThinks/MaiaM Alchemist preprint, companion to this work.
- Zellers, R. et al. (2019). *HellaSwag: Can a Machine Really Finish Your Sentence?* ACL.

Appendix A — Sample of the 256-cell catalog

The full catalog is generated by `packages/aamt-foundations/scripts/seed_oracle_cards.py` and stored at `packages/aamt-foundations/oracle-256-cards.json`. Each cell carries an internal id (“upper-lower”), a kind tag (**resonance** for the 16 diagonal cells, **transition** for the 240 off-diagonal cells), upper and lower Meji metadata (numeric id, AAMT-native name, Ifá lineage name), and a small set of stratification- relevant descriptors (title, tagline, key question).

A representative sample of 8 cells (2 resonance, 6 transition):

id	kind	upper	lower	popcount	stage
0-0	resonance	Apex	Apex	8	8
1-1	resonance	Hollow	Hollow	0	0
0-1	transition	Apex	Hollow	4	4
1-0	transition	Hollow	Apex	4	4
2-13	transition	Heart-Mind	Lens	3	3
13-2	transition	Lens	Heart-Mind	3	3
6-11	transition	Compass	Tide	3	3
4-5	transition	Bloom	Root	4	4

Note: (0-1) and (1-0) are distinct cells with identical popcount and stage but opposite *direction* (which mask is upper, which is lower). For curricula that do not care about direction, a symmetric variant collapsing each off-diagonal pair into a single bucket yields a 136- cell taxonomy (16 diagonals + $(16 \times 15)/2 = 120$ unordered pairs). We retain the directional form by default because

the upper/lower asymmetry encodes the F-phase/M-phase distinction used by the breath-cycle operators of the AAMT Foundations.

Appendix B — Reference Implementation

Key files:

- **Meji frame and metadata:** `packages/aamt-foundations/yare-vortex-mathematics.json` — declarative spec of the 16 Meji, the 256 Odu structure, and the TERA-to-line mapping.
- **256-cell catalog:** `packages/aamt-foundations/oracle-256-cards.json` (generated) and the generator `scripts/seed_oracle_cards.py`. The generator emits 16 resonance cards + 240 transition cards with deterministic composition from per-archetype voice profiles.
- **Soft Meji distribution runtime:** `packages/vortex-runtime/src/index.ts` exports the `mejiDistribution(tera)` and `sampleFromDist(dist)` primitives. A duplicated copy lives in `apps/oracle/lib/odu-data.ts` for bundler-isolation reasons; the canonical implementation is the runtime package.
- **Data-bucketing protocol:** `maiam-chemist/packages/code-corpus/` produces per-example TERA labels and per-cell jsonl shards; the curriculum loader consumes these shards in stage order.
- **Curriculum loader:** `maiam-chemist/packages/training-runs/training_runs/run_recorder.py` logs per-cell loss traces; the curriculum-aware sampler is configured per training run.

Reference checkpoints will be released alongside v1.1 of this preprint.

End of preprint v1.0.